

رحمة

RAHMA

AI-Powered Guardian Application for Vulnerable Populations
United Arab Emirates

Official Innovation Proposal

Prepared for: UAE Ministry of Community Development

In partnership with: Ministry of Health & Prevention UAE

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Executive Summary

Rahma (رحمة — Arabic for 'compassion') is a first-of-its-kind AI guardian application built for the UAE. It is designed to protect and serve three of society's most vulnerable groups: the elderly, children, and patients with special or chronic needs.

The app operates with a single tap — no complex navigation, no typing required. It connects users instantly to emergency services, hospitals, pharmacies, and family caregivers, while intelligently adapting to their language, situation, and medical profile.

Rahma is not just an app — it is a digital companion that gives vulnerable people a voice, security, and access to care at the speed of life. Nothing like it exists anywhere in the world.

1 Touch to activate any service	12+ languages supported	3 target user populations	24/7 AI-powered assistance
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Problem Statement

Millions of vulnerable people in the UAE face daily challenges that existing apps are not designed to solve. They need help that adapts to them — not the other way around. Current digital health and emergency solutions share a critical failure: they are built for tech-savvy adults in full health.

1. Elderly Population (800,000+ residents)

Elderly residents face limited digital literacy, a high risk of falls and medical emergencies, language barriers — especially among expatriate elderly — complex medication schedules, and social isolation. Existing apps require multiple menu navigations and text input that is simply not accessible to this group.

2. Children (1.2 million+ in UAE)

Children face risks when alone — accidents, getting lost, stranger danger, and health emergencies at school or in public. They need to be able to contact trusted adults instantly without the complexity of a phone call or understanding English app interfaces.

3. Patients with Special Needs (100,000+ registered)

Chronic patients, individuals with physical or neurological disabilities, and those with conditions affecting communication often cannot describe symptoms clearly, navigate phone trees during an emergency, or manage medication independently. Current solutions are fragmented across multiple apps with no unified emergency response.

The Core Gap

No application in the UAE or globally combines emergency dispatch, medicine ordering, hospital finding, multilingual AI triage, child safety, and wearable integration into a single, one-touch interface designed specifically for all three vulnerable groups simultaneously.

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The One-Touch Experience

The entire Rahma interface is designed around a single large activation button — prominent, impossible to miss, and impossible to activate accidentally. The button fills the majority of the phone screen. Users press and hold for two seconds, and the AI takes over from there.

How It Works — Step by Step

- User presses the large Rahma button (or it auto-activates via fall detection)
- AI identifies language automatically from the user's first sound or configured profile
- AI asks a single, simple voice question: 'Are you hurt? Need medicine? Feeling unwell? Lost?'
- Based on the response — or silence — the correct service is activated in under 10 seconds
- Caregivers and family members receive an instant notification with the user's live GPS location

Design Principle

Every design decision in Rahma is tested against one rule: Can a 75-year-old who has never used a smartphone, does not speak English, and is in distress use this feature in under 15 seconds? If not, it is redesigned.

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Core Features

Every feature is built around zero-barrier access. Nothing requires typing or complex navigation. All interactions are voice-first with visual and haptic fallbacks.

Feature	Description
Emergency Dispatch	One tap sends GPS location and user medical profile directly to UAE emergency services (998/999). AI provides pre-arrival first aid instructions to bystanders in the correct language.
Medicine Ordering	Voice-activated prescription ordering from registered UAE pharmacies. AI reminds users of doses using sound, vibration, and light signals — no typing required.
Hospital Finder	Instantly locates and connects with the nearest hospital or specialist clinic. Pre-fills patient details from user profile to speed up registration on arrival.
AI Health Triage	Conversational AI assesses symptoms using voice and guides users on whether to call an ambulance, visit a clinic, or apply first aid — in their language.
Guardian Alerts	Family members and caregivers receive instant push notifications and live location when the app is activated. A silent panic mode alerts without user sound.
Multilingual Voice AI	Understands and responds in 12+ languages — adapting automatically to the speaker without any language selection needed.
Safe Zones & Geofencing	Caregivers define safe zones for children and elderly users. Instant alerts when a user leaves or enters a zone. Historical route tracking available.
Wearable Integration	Connects with smartwatches and fall detectors. Auto-triggers emergency mode if a fall or abnormal heartrate is detected without user input.

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Multilingual Support

The UAE is home to over 200 nationalities. Rahma speaks their language — automatically, without requiring the user to select a language preference. The AI detects language from the first few words spoken and switches instantly.

Languages supported at launch:

- Arabic (Modern Standard + Gulf dialect)
- English
- Hindi and Urdu
- Tagalog
- Bengali
- Malayalam and Tamil
- Persian (Farsi)
- Russian
- French
- Nepali

Post-launch languages roadmap includes Sinhala, Amharic, Swahili, and Pashto.

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What Makes Rahma Unprecedented

A thorough analysis of global app stores and healthcare technology databases confirms no existing application combines all of the following in a single, unified, one-touch interface for vulnerable populations:

1. Zero-UI Emergency AI

All existing emergency apps — including those in the US, UK, and GCC — still require the user to choose an action (call ambulance, call police, call poison control, etc.). Rahma's AI makes that decision through a brief voice exchange, removing the cognitive burden from distressed users entirely.

2. Cross-Group Unified Design

Apps exist for elderly care OR child safety OR patient tracking — never all three within a single, adaptive interface that resizes, simplifies, and adjusts based on who is pressing the button. Rahma uses AI-identified user profiles to serve all three groups from one app.

3. UAE-Native AI Intelligence

Rahma's AI model is trained specifically on UAE healthcare infrastructure, Arabic dialects, local emergency dispatch protocols, UAE Ministry of Health APIs, and UAE pharmacy networks — not a generic global model repurposed for the region.

4. Passive Emergency Detection

Through wearable integration and phone motion sensors, Rahma can detect a fall, extended immobility, or a distress cry and activate emergency protocols automatically — without any button press from the user.

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Technical Architecture

Rahma is built on a hybrid on-device and cloud AI architecture, ensuring emergency functionality is never dependent on internet connectivity.

Component	Technology / Integration
AI Core	Multilingual LLM with Arabic dialect training, on-device + cloud hybrid processing
Voice Recognition	Real-time speech-to-intent, dialect-aware, works offline
Emergency Dispatch	UAE National Ambulance (998/999), Dubai Police, ADPOC APIs
Healthcare Records	Ministry of Health EMR, Malaffi (Abu Dhabi), DHA (Dubai)
Pharmacy Network	Aster Pharmacy, Life Pharmacy, Boots UAE, government pharmacy APIs
Mobile Platform	iOS and Android, smartwatch companion apps, WhatsApp emergency channel
Data Security	UAE NESAC compliant, end-to-end encrypted, health data stored on-premise in UAE
Offline Mode	Core emergency function operates with zero internet via SMS fallback

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Phased Implementation Roadmap

Phase 1 — Q3 2026: Pilot Launch (Dubai & Abu Dhabi)

- Deploy to 5,000 registered elderly and special needs patients via DHA and DoH
- One-touch emergency dispatch and hospital finder live
- Arabic and English language support
- Government beta testing and clinical validation

Phase 2 — Q4 2026: Children's Module & Multilingual Expansion

- Launch child safety features with school integration and geofencing
- Expand to 10 languages
- Pharmacy ordering and medicine reminder system live
- Target: 50,000 active users

Phase 3 — Q2 2027: UAE-Wide Rollout

- Full public launch across all 7 emirates
- Smartwatch and fall-detector hardware integration
- AI health triage system live
- Target: 250,000 active users

Phase 4 — Q4 2027: GCC Expansion

- Expand to Saudi Arabia, Qatar, and Bahrain under a GCC partnership framework
- Launch Rahma Companion — a proactive AI that checks in on users daily
- Behavioural decline detection using longitudinal AI health models

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Call to Action

Rahma is ready to move from proposal to pilot. We invite the UAE Ministry of Community Development and Ministry of Health & Prevention to co-develop this vision into reality under the umbrella of UAE's 2031 National Strategy for People of Determination and the broader ambitions of UAE Centennial 2071.

This is a proposal for a system that does not yet exist anywhere in the world. The UAE has the infrastructure, the vision, the regulatory environment, and the humanitarian mandate to build it first.

Our Commitment

A more compassionate UAE starts here. Rahma will give a voice, security, and access to care to those who need it most — the elderly, the children, and the patients — with the dignity they deserve.

رحمة — رحمة للجميع
Rahma — Compassion for All